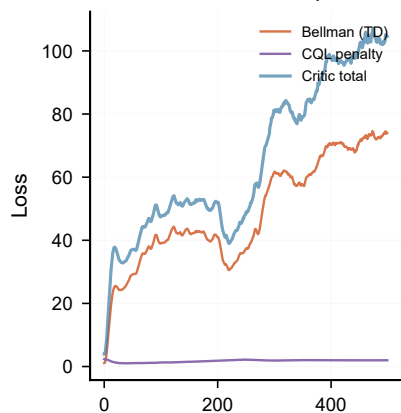
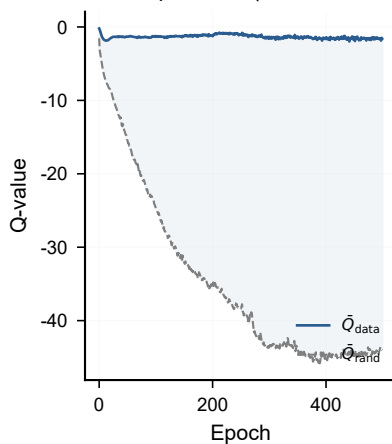


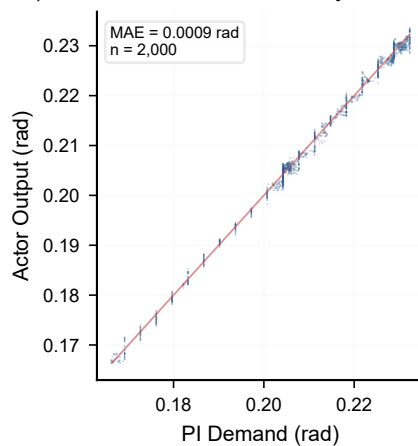
a Critic Loss Decomposition



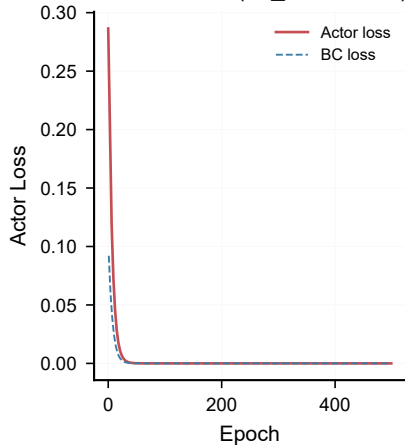
b Q-value Separation (CQL Conservatism)



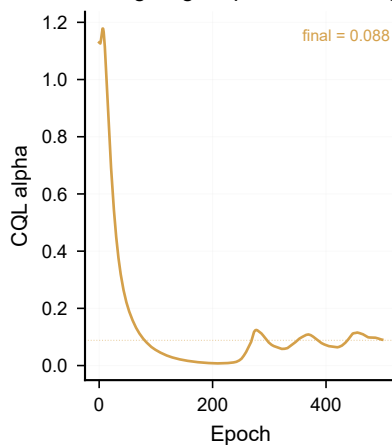
c Action Fidelity



d Actor Loss (-Q_norm + BC)



e Lagrange alpha Auto-tuning



f Final Metrics

Q_{data}	-1.60
Q_{rand}	-44.03
Delta Q gap	42.4
BC loss	1.5e-06
CQL alpha	0.088
Bellman loss	74.4
MAE (rad)	0.0009

CQL working: Delta Q = 42, Actor fidelity MAE = 0.0009 rad